

Modular Program Hierarchy

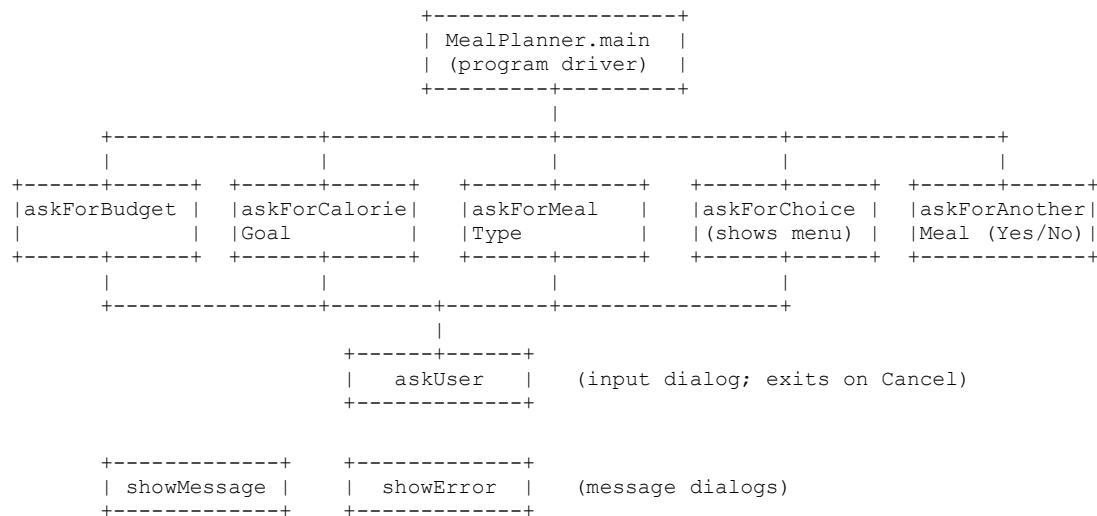
Justin Pence, Sammy Kahingo, Asim Adam, Rohit Kumar

IT 106 - DL1

GMU Campus Budget & Calorie Meal Planner (Java)

Hierarchy Chart

The program is a single class, MealPlanner. Its modules are the main method (the driver) and a set of helper methods, each with one job. main calls the input methods to collect choices, and every input method uses askUser to show a JOptionPane input dialog. The output methods showMessage and showError display JOptionPane message dialogs.



main calls showMessage for the welcome, meal confirmations, stop messages, and the daily summary. The ask-methods call showError when input is invalid.

Shared data used by main: menu arrays (names, costs, calories for breakfast, lunch, dinner) and constants (FEMALE_GOAL, MALE_GOAL, CALORIES_PER_MILE, CHEAPEST_MEAL, TITLE).

Module Roles

Module	Type	Role	Returns / Output
main	Driver	Runs setup, the meal loop, and the daily summary. Updates the budget and calorie total.	Shows meal confirmations and the final summary
Menu arrays	Data	Parallel arrays of names, costs, and calories for breakfast, lunch, and dinner.	Read by main to build the menu

Module	Type	Role	Returns / Output
Constants	Data	Calorie goals, calories per mile, cheapest meal cost, dialog title.	Read by main and helpers
askForBudget	Input	Gets a positive number for the daily budget.	double budget
askForCalorieGoal	Input	Gets M or F and picks the matching calorie goal.	int goal (2000 or 2500)
askForMealType	Input	Gets B, L, or D.	"breakfast", "lunch", or "dinner"
askForChoice	Input	Shows the menu and gets 1, 2, or 3.	int choice
askForAnotherMeal	Input	Yes/No confirm dialog.	boolean
askUser	Input helper	Shows a JOptionPane input dialog; exits cleanly on Cancel.	trimmed String
showMessage	Output helper	Shows an information dialog.	Dialog box
showError	Output helper	Shows an error dialog for invalid input.	Dialog box

Module Interactions

1. Setup

1. main calls showMessage to welcome the user.
2. main calls askForBudget, which uses askUser and shows showError until a positive number is entered.
3. main calls askForCalorieGoal, which returns 2000 or 2500, then shows the goal with showMessage.
4. main compares the budget to CHEAPEST_MEAL. If no meal is affordable, it says so and skips the meal loop.

2. Meal Loop

1. askForMealType returns breakfast, lunch, or dinner, and main selects the matching menu arrays.
2. main builds the menu text from the arrays and passes it to askForChoice, which returns 1-3 (converted to index 0-2).
3. If the meal costs more than the remaining budget, main shows a message and the loop repeats.
4. Otherwise main subtracts the cost, adds the calories, and shows a confirmation with showMessage.

5. The loop ends if the budget is below CHEAPEST_MEAL, the calorie goal is reached, or askForAnotherMeal returns false.

3. Summary

1. main compares total calories to the goal and builds one summary message.
2. If the user is over the goal, main calculates jogging miles (extra calories / CALORIES_PER_MILE) and adds the suggestion.
3. main shows the summary with showMessage and the program ends.

Data Flow Summary

From	Data	To
askForBudget	Daily budget	main (budget variable)
askForCalorieGoal	Calorie goal	main (goal variable)
askForMealType	Meal type	main (selects menu arrays)
Menu arrays	Names, costs, calories	main (builds menu text)
askForChoice	Menu choice	main (array index)
main	Updated budget and calorie total	showMessage (confirmation)
askForAnotherMeal	Yes / No	main (loop control)
main	Totals, goal, jogging miles	showMessage (daily summary)